



Research Article

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The Impact of GenAI-Automatic Corrective Feedback (ACF) Integration in Collaborative Writing Based on Metacognitive Instruction on Argumentative Essay Writing Skills

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Abstract

The purpose of this study was to investigate the impact of GenAI-Automatic Corrective Feedback (ACF) integration in collaborative writing based on metacognitive instruction on argumentative essay writing skills. The method used in this study was a quasi-experimental study involving 250 students divided into two groups: an experimental group and a control group. The experimental group received the GenAI-Automatic Corrective Feedback (ACF) integration intervention in collaborative writing based on metacognitive instruction, while the control group used a genre-based writing approach. The instruments used included writing assessments covering lexical, accuracy, and fluency. Data analysis used ANOVA and regression analysis to investigate the impact of the intervention on writing skills. The research findings showed that the GenAI-Automatic Corrective Feedback (ACF) intervention in collaborative writing based on metacognitive instruction significantly improved argumentative essay writing skills compared to the genre-based writing intervention. Improvements in argumentative essay writing skills in the experimental group were evident in several aspects, including lexical complexity (lexical density, lexical sophistication, and lexical variety), accuracy, and fluency. Increased lexical complexity was evident in the use of complex vocabulary, complex sentence combinations, and more varied lexical use. Improved argumentative essay accuracy was evident in the accuracy of sentences and paragraphs and supporting ideas relevant to the problem of the argumentative essay. Improved argumentative essay fluency was evident in the use of sentences and grammar with minimal errors and no discordant sentences. The components of metacognitive instruction that significantly contributed to argumentative essay writing skills were metacognitive knowledge, including declarative, procedural, and conditional knowledge, and metacognitive regulation, including planning, monitoring, and evaluation, as well as information management and debugging strategies.

Keywords: Collaborative writing approach, metacognitive instruction, writing skills development.

1. Introduction

Writing skills encompass a multifaceted activity that integrates various cognitive, metacognitive, linguistic, emotional, and motivational processes during the writing journey. Each of these processes is recursive, necessitating writers to plan, translate, and evaluate both the writing process and the end result (Benabbes et al., 2025; Farahian & Avarzamani, 2018). This idea highlights the important function of metacognition in the writing process. The significance of metacognition is evident in the idea of writing as a form of applied metacognition. Metacognition involves recognizing and comprehending one's cognitive activities, enabling the writer to oversee and adjust the writing process to fulfill established writing standards (Jmaiel et al., 2025; Lee & Bosch, 2025).

Learners' writing success is not solely determined by the teaching model but also by the development of learners' metacognitive abilities to regulate themselves during the learning process. This concept is reinforced by several previous studies showing a strong correlation between the role of metacognition and writing ability (Choo & Double, 2025; Pousi et al., 2025). However, in these previous studies, the mechanisms of metacognitive instruction in the context of collaborative writing have not been explored more comprehensively.

Metacognitive abilities in writing will be more optimally facilitated by the use of technology, one of which is GenAI. GenAI- Automatic Corrective Feedback (ACF) can facilitate students in optimizing metacognitive abilities in the context of individual and collaborative writing. The development of technology use in language learning can optimize language competence. Writing is seen as a multifaceted cognitive process involving learners, teachers, tasks, and media that contribute to the formation of meaning in the language learning process (Fono & Zohar, 2024; Mohammed & Khalid, 2025).

Several previous studies have found that metacognitive learning and collaborative writing approaches have a significant impact on students' writing abilities (Khosravi et al., 2023; Kim et al., 2024). Furthermore, the findings of other studies also confirm that metacognitive instruction facilitates writers to improve writing skills, choose writing strategies, increase student autonomy, and use strategies in different contexts (Mallahi, 2024; Rezai et al., 2023).

Furthermore, the collaborative writing approach is able to create texts with collaborative responsibilities that have potential in language learning, especially in improving writing skills. This approach encourages interaction, filling gaps in understanding, feedback, and quality output (Feng et al., 2024; Kim et al., 2024). This collaborative writing process is not limited to the abilities of one individual alone but is complementary, there is social interaction, and improves the quality of writing.

Several previous studies have investigated metacognitive instruction and technology use. One study revealed that metacognition-based argumentative writing instruction improved argument organization (Anggraeni et al., 2025; Mekheimer, 2025). Furthermore, another study demonstrated that metacognition-based GenAI-Coursera integration improved students' writing skills (Jin, 2025; Khuder & Negretti, 2025).

Another study revealed that the writerpreneur platform not only improved essay writing skills but also leveraged writing opportunities in the entrepreneurial field (Österholm et al., 2025; Taufiquilloh et al., 2025). Another study revealed that the use of webquest technology improved students' writing and critical thinking skills (Paulsen & Davidsen, 2025; Usher et al., 2024). The current research differs from previous studies. The writing measurements used in this study are more complex, encompassing complexity, accuracy, and fluency. Furthermore, the metacognitive instruction in this study was conducted in the context of collaborative writing to develop argumentative essay writing skills. Based on these explanations, the researchers formulated the following research questions:

- a) What impact does the integration of GenAI-Automatic Corrective Feedback in

collaborative writing based on metacognitive instruction have on argumentative essay writing skills?

- b) What components of metacognitive instruction contribute significantly to argumentative essay writing skills?

2. Literature Review

2.1 Collaborative Writing Approach in Writing Learning

Collaborative writing instruction is characterized by interactive activities between learners, instructors, and texts, all working together to construct meaning. This collaborative writing approach is shaped by sociocultural theory, which views teaching as an interactive pedagogical activity designed to facilitate collective scaffolding, discussion, support, and social interaction throughout the writing process (Fono & Zohar, 2024; Moritz et al., 2022). Cognitive concepts suggest that the collaborative writing process encourages learners to engage actively in different learning tasks, negotiate meanings, connect ideas, and both give and receive feedback to achieve high-quality outcomes.

Earlier research indicates that collaborative writing encourages positive perceptions, enhances the quality of the experience, and offers feedback from learners with deeper comprehension (Creative et al., 2025; Fraiwan et al., 2025). Furthermore, other studies have reported that collaborative writing produces better writing quality than individual writing in the posttest phase.

Several experts have agreed that collaborative writing assessments are based on linguistic aspects of complexity, accuracy, and fluency. Other studies have confirmed that collaborative writing has a more lasting impact on writing ability than individual writing. This finding is consistent across writing experiences across various genres (Jeliseh & Xodabande, 2025; Shen & Bai, 2025).

The benefits of collaborative writing interventions can be transferred to individual writing skills, which can be assessed using a rubric covering content, grammar, vocabulary, and mechanics. Previous studies revealed that collaborative writing interventions only had a significant impact on fluency and accuracy, but not on writing complexity (Anthonysamy et al., 2025; Creative et al., 2025). This is because collaborative interventions focused solely on grammatical accuracy.

The inconsistent findings from several studies on collaborative writing outcomes have prompted researchers to investigate the suboptimal impacts of collaborative writing (Wang & Wang, 2025; Zhu et al., 2025). Along with the development of writing instruction, researchers have proposed a metacognitive approach to facilitate learners' optimal collaborative writing. This metacognitive approach can bridge self-efficacy, writing strategies, and writing collaboration to achieve better writing quality.

This idea is also supported by previous studies showing that collaborative metacognitive knowledge is strongly correlated with collaborative writing, interaction patterns, and language proficiency improvement (Seban et al., 2025; Zhang & Jun, 2025). Although previous studies have shown that collaborative writing metacognitive knowledge is significantly correlated with interaction processes during writing, these studies have not yet investigated the relationship between metacognition-based collaborative writing and writing outcomes.

2.2 Metacognitive Instruction in Writing Learning

Writing instruction can facilitate learners' use of effective writing strategies, improve writing skills, and develop learner potential. Metacognitive instruction requires interaction between

components such as the instructor, the learner, and the writing task. This argument reinforces the need for deeper exploration of the relationship between metacognitive instruction and collaborative writing approaches (Eduardo et al., 2025; Ho & Lau, 2025).

A recent study integrated metacognition and collaborative writing. The results showed that metacognitive-based collaborative writing only had a significant impact at the secondary school level (Liu & Zhang, 2025; Zhan & Teng, 2025). However, at lower school levels, it did not have a significant impact. Another study revealed that metacognitive instruction in individual writing learning was suboptimal due to the lack of peer feedback (Paulsen & Davidsen, 2025; Usher et al., 2024). Metacognitive instruction in the context of collaborative writing learning is considered effective because peers can provide feedback to each other, are able to negotiate form and meaning, complement each other's shortcomings in the writing process, so that they are able to produce better quality written products.

Cognitive and metacognitive factors play a crucial role in supporting the effectiveness of collaborative writing in language learning. Cognitive aspects contribute when learners actively participate in each activity, developing writing skills and gaining a deeper understanding of collaborative writing (Khuder & Negretti, 2025; Österholm et al., 2025).

Although cognitive activities support collaborative writing, metacognitive processes are still necessary in planning, monitoring, and evaluating these cognitive activities and the resulting written products. To optimize collaborative writing interventions in language learning, metacognitive instruction must be appropriately designed to encourage higher-order thinking skills so that learners are more optimal in the writing process (Jin, 2025; Mekheimer, 2025).

Prior research compared various instructional strategies to examine how metacognitive education in collaborative writing affected the growth of academic writing abilities (Carmona et al., 2025; Garcés-Manzanera, 2024). These methods included individual writing, collaborative writing without metacognitive training, metacognitive instruction with collaborative writing, and metacognitive instruction without collaborative writing.

These studies revealed that academic writing skills are successfully enhanced by metacognitive teaching, which includes knowledge (tasks, techniques, and self-efficacy) and metacognitive regulation (planning, monitoring, and assessing). The study's findings demonstrate how metacognitive teaching via group writing can promote advanced information processing, reflect on the writing process, and eventually enhance students' academic writing abilities (Gadd & Parr, 2025; Rezai et al., 2023). Researchers conducted this study to better investigate the influence of incorporating metacognitive learning into collaborative writing because there aren't many studies on this topic.

2.3 GenAI - Auto-Correction Feedback (ACF) in Writing Learning

Auto-Correction Feedback (ACF) is a GenAI that can be used as a scaffolding tool to provide corrective feedback to improve writing quality. Current writing studies focus on the use of AI and other technologies adopted in writing instruction (Kim et al., 2024; ten Peze et al., 2024). ACF is used in the writing learning process to help students address issues related to aspects that can improve writing quality, such as grammar, punctuation, spelling, and conventions, while also providing time for revision (Mohammed & Khalid, 2025; Rahimi et al., 2024).

Students also receive appropriate and consistent metalinguistic explanations, which can facilitate students' improvement in writing accuracy and help teachers anticipate time constraints. Feedback on writing errors also facilitates students' focus on errors, allowing for effective correction. Several previous studies have explored various online corrective feedback platforms (Fraivan et al., 2025; Moritz et al., 2022).

Previous research has shown that automated feedback using the ACF online platform significantly contributes to essay writing skills in terms of organizing ideas and using grammar

(Fraivan et al., 2025; Wang & Wang, 2025). Furthermore, other research has revealed that the Write & Improve online platform is able to accurately diagnose students' writing abilities according to the Common European Framework of Reference (CEFR) scale levels and is able to facilitate students in completing essay, report, and argument writing tasks with good quality (Eduardo et al., 2025; Ka et al., 2025).

Furthermore, other research has revealed that the ACF platform is able to significantly contribute to grammatical accuracy and other dimensions of writing performance such as task achievement, cohesion and coherence, and lexicon, as well as active student participation (Liu & Zhang, 2025; Parker et al., 2025).

3. Method

3.1 Design and Participants

This quasi-experimental study investigated the impact of a GenAI-Automatic Corrective Feedback (ACF) intervention integrating collaborative writing based on metacognitive instruction on argumentative essay writing skills. The study involved 250 university students in their second-seventh semesters, aged 17-19.

The participants were 45% male and 55% female. Participants were divided into two groups: an experimental group and a control group. The participants were randomly assigned to each group. The experimental group received a collaborative writing approach based on metacognitive instruction, while the control group received genre-based writing instruction. All participants completed a consent form, ensuring voluntary participation.

3.2 Instruments

This study employed several instruments, including the types of writing intervention tasks and an argumentative writing ability assessment. The three argumentative writing tasks used in the experimental group were used. The argumentative genre was chosen because this genre forms the basis for students' competency in writing scientific papers and final assignments, which are required at the university level. Improving the ability to write argumentative essays can motivate students to complete all coursework during the lecture process.

The argumentative essay writing test was adapted from the IELTS which was tested in the pretest and posttest phases. The duration of writing an argumentative essay was 45 minutes to complete. The subsequent tool employed in this research was a metric for assessing argumentative essays, focusing on dimensions of lexical complexity, accuracy, and fluency. This metric was employed to evaluate written language skills, language proficiency, and language development as adapted from Lu (2012).

The complexity dimension pertains to lexical richness, intricacy, and variety in vocabulary. Lexical density is evaluated by comparing the count of lexical words to the overall total of words. Lexical sophistication is measured by comparing the count of sophisticated lexical terms to the overall count of lexical terms. Additionally, lexical diversity is evaluated based on the proportion of adjusted type tokens. The complexity assessment of the analysis utilized the Lexical Complexity Analyzer application tool.

The following element is precision. Accuracy pertains to how closely the language employed conforms to the standards of language rules. The accuracy aspect pertains to a person's capability to write in a fluid and cohesive manner without any hindrances. Accuracy is evaluated based on the proportion of correct clauses, whereas fluency is measured by the total words produced within a specific timeframe. The evaluations of accuracy and fluency utilize Shea's (2014) framework.

The reliability of the task used to assess writing ability was assessed through expert judgment involving five assessors with a PhD in linguistics. Intraclass correlation coefficient analysis was used to investigate inter-rater reliability. The analysis showed a reliability of 0.84, indicating that the assessment instrument met the reliability criteria and was therefore suitable for use. Next, the assessment metric instrument was empirically evaluated on 50 students to determine its validity. Pearson correlation was used because it is a simple technique for evaluating the level of linear correlation between questions.

The assessment results showed an average Pearson correlation coefficient of 0.86 for each assessment item. Furthermore, a split-half approach was used to evaluate the instrument's reliability. The evaluation showed a reliability level of 0.83. Based on the validity and analysis results, this instrument met the requirements for use in this study.

3.3 GenAI-ACF Intervention in Collaborative Writing Based on Metacognitive Instruction

The GenAI-ACF intervention in collaborative writing based on metacognitive instruction was used to enhance metacognitive knowledge and regulation to support the development of students' argumentative essay writing skills. The intervention training encompassed several components, such as metacognitive knowledge, including declarative knowledge (explaining topics), procedural knowledge (explaining procedures), and conditional knowledge (examples of good argumentative essays), as well as metacognitive regulation, including planning, monitoring, evaluation, information management strategies, and debugging strategies.

Students were trained to produce high-quality argumentative essays through collaborative interventions and metacognitive instruction. The collaborative writing learning based on metacognitive instruction was implemented for one semester in the experimental group, while the control group received a genre-based writing intervention with conventional writing learning for the same duration.

Table 1. Writing stages with GenAI-ACF in collaborative writing based on metacognitive instruction.

Stages	Explanation
Presentation of metacognitive knowledge	In this stage, students are provided with metacognitive knowledge related to writing argumentative essays and using GenAI-ACF.
Declarative knowledge	The instructor presents the concept of understanding argumentative essays, including the definition, organization, and paragraph development strategies, as well as the corrective operations of GenAI-ACF.
Procedural knowledge	The instructor explains the stages of writing an argumentative essay, including topic determination, development, and organization of each paragraph.
Conditional knowledge	The instructor provides examples of argumentative essays for comprehensive understanding through collaborative discussions to complement each other's understanding of essays and GenAI-ACF.
Metacognitive regulation training	In this stage, the instructor provides instructions on metacognitive regulation in writing argumentative essays.
Planning	Students plan argumentative essay topics based on the given topics. Argumentative essay topics include natural disasters, social life, and technological developments.
Monitoring	Students are instructed to write argumentative essays in pairs, dividing the structure of an argumentative essay. At this stage, the teacher and students jointly monitor and check writing errors using GenAI-ACF.

Evaluation	Students are instructed to evaluate their writing based on the GenAI-ACF checks and discuss errors in pairs, covering aspects of lexical complexity, accuracy, fluency, and grammar.
Information management strategy	Students are trained to understand, organize, elaborate, and effectively use their understanding of argumentative essays.
Debugging strategy	The final stage is when students are asked to correct their errors in writing the argumentative essay based on the evaluation results.

3.4 Procedure

Before the intervention, all participants received a detailed explanation of the study and completed a consent form to participate. Initially, all groups received a pretest on argumentative essay writing to assess their initial abilities. After ensuring that all participants in both groups were equal, the intervention was administered to each group. Participants were paired in the experimental group to receive a writing intervention using GenAI-ACF in collaborative writing based on metacognitive instruction, while the control group received a conventional, genre-based writing intervention.

Intervention sessions were conducted three times a week, each lasting 50 minutes. The collaborative writing intervention used three argumentative essays as examples and provided assignments. The instructor facilitated the intervention only when students needed assistance or had difficulty. In the control group, students received a genre-based writing intervention with conventional instruction. After all interventions, both groups received an argumentative essay writing assignment in the posttest and a delayed posttest seven weeks after the intervention ended to investigate argumentative essay writing skills and the long-term impact of the intervention.

3.5 Data Analysis

The data analysis used in this study included descriptive analysis and a mixed ANOVA to investigate the differences in the intervention's impact on argumentative essay writing ability between the two groups. Descriptive statistical analysis was conducted on each aspect in detail, including lexical complexity, accuracy, and fluency.

ANOVA tests were conducted to investigate main effects and interaction effects between time and group, and paired analyses were conducted at each phase to determine differences in argumentative essay writing ability at each phase. Additionally, a mixed between-subjects ANOVA was also conducted to examine simple effects within and between subjects.

4. Results

The first analysis was conducted on the lexical complexity of the intervention results for both groups. The results of the descriptive statistical analysis of the lexical complexity of argumentative essays across the three exams are presented in Table 2. The analysis revealed improvements in both groups.

However, the experimental group showed greater improvement in lexical density, lexical sophistication, and lexical variety. Further analysis was conducted using mixed ANOVA inferential statistics for each aspect. First, the results of the mixed ANOVA on lexical density showed that the intervention had a significant main effect of time with an $F(2, 60) = 17.925$, $p < 0.001$, and Partial $\eta^2 = 0.573$. The main effect of group was quite significant with a value of $F(1, 65) = 0.135$, $p = 0.845$, Partial $\eta^2 = 0.003$, and a significant interaction effect of time and group with a value of $F(2, 60) = 0.185$, $p = 0.836$, Partial $\eta^2 = 0.007$.

Furthermore, the results of the pairwise analysis found that both groups showed an increase in lexical density, with the experimental group showing a better increase in each phase. This increase was maintained until the delayed posttest phase. Another finding from the intersubject comparison analysis was that the experimental group showed a better increase in lexical density than the control group. From these findings, it can be concluded that conventional writing instruction is also able to increase lexical density, but not better than the collaborative writing intervention based on metacognitive instruction.

Table 2. Descriptive statistics of lexical complexity.

Lexical complexity	Group	N	Pre-test		Post-test		Delayed posttest	
			M	SD	M	SD	M	SD
lexical density (Nlex/N)	Experiment	12	5.52	.0	7.7	.03	8.8	.0
		5	5	46	84	0	43	45
	Control	12	5.6	.0	5.6	.0	5.55	.03
		5	27	48	68	46	8	0
lexical sophistication (Nslex/Nlex)	Experiment	12	5.11	.05	8.16	.03	8.4	.0
		5	7	3	5	8	21	48
	Control	12	5.21	.0	5.57	.0	5.25	.05
		5	4	42	3	45	4	0
lexical variation (CTTR)	Experiment	12	4.5	.45	7.4	.6	8.4	.42
		5	67	8	68	40	76	5
	Control	12	4.6	.47	5.72	.43	5.7	.35
		5	78	8	6	5	83	6

Nlex/N number of lexical words to the total number of words, Nslex/Nlex number of sophisticated lexical words to the total number of lexical words, CTTR corrected type token ratio

The next analysis was carried out using a mixed ANOVA test on the lexical sophistication aspect. The results of the analysis showed that there was a significant main effect of the time variable with a value of $F(2, 60) = 105.785$, $p < 0.001$, η^2 Partial = 0.658. Also found was the main effect of the group with a value of $F(1, 60) = 5.025$, $p = 0.063$, η^2 Partial = 0.074 and a significant interaction effect between time and group with a value of $F(2, 60) = 0.001$, $p = 0.867$, η^2 Partial = 0.001.

Furthermore, the pairwise analysis found a consistent increase in lexical sophistication in the following pairwise analyses. Another finding was a significant difference in lexical sophistication at each phase between the two groups. Therefore, it can be concluded that the experimental group's intervention through collaborative writing based on metacognitive instruction was more effective in improving the lexical sophistication of argumentative essay writing than the genre-based writing intervention in the control group.

The next analysis was a mixed ANOVA test on lexical variation. The results of the analysis showed that there was a significant main effect of the time variable with a value of $F(2, 60) = 21.053$, $p < 0.001$, η^2 Partial = 0.463. In addition, a significant main effect of the group was also found with a value of $F(1, 60) = 8.175$, $p = 0.010$, η^2 Partial = 0.110 and a fairly significant interaction effect of the interaction of time and group with a value of $F(2, 60) = 6.768$, $p = 0.007$, η^2 Partial

= 0.174 on the performance of lexical variation in student writing.

Furthermore, based on the results of the pairwise analysis, it was found that students in the experimental group were able to increase lexical variation from each phase (pre- vs. post-test, $p < .001$; pre- vs. delayed post-test, $p < .001$) to be able to maintain their competence until the delayed post-test phase (post- vs. delayed post-test, $p > .05$). However, students in the control group's lexical variation performance remained stable and did not show a significant increase in all three phases (pre-post-test, $p > .05$; pre- vs. delayed post-test, $p > .05$). Furthermore, the results of the intersubject comparison analysis showed that the lexical variation performance of both groups was equivalent in the pretest phase ($p > .05$).

However, in the posttest phase, students in the experimental group showed a more significant increase than the control group with a p value = .017 with a small effect size, Cohen's $d = .568$, but in the delayed posttest phase, the difference was not significant, $p > .05$.

The next analysis was the accuracy of students' argumentative essays. The descriptive statistical analysis of argumentative essay accuracy is presented in Table 3. Based on the analysis, both groups showed an increase in accuracy scores in the posttest phase, but a more significant increase was found in the experimental group.

Next, an inferential statistical analysis using a mixed ANOVA test was conducted to investigate the impact of the intervention in more detail. Based on the results of the mixed ANOVA test on accuracy performance, a significant main effect of time was found with a value of <0.001 , $F(2, 60) = 12.568$, $p < 0.001$, Partial $\eta^2 = 0.384$. Furthermore, a significant main effect of group was also found with a value of <0.001 , $F(1, 60) = 5.754$, $p = 0.037$, Partial $\eta^2 = 0.082$. However, the interaction effect between time and group was not significant with a value of <0.001 , $F(2, 60) = 3.586$, $p = 0.100$, Partial $\eta^2 = 0.083$.

Table 3. Descriptive statistics of accuracy of argumentative essays.

Accuracy	Group	N	Pre-test		Post-test		Delayed posttest	
			M	SD	M	SD	M	SD
EFC/C	Experiment	12	5.6	.17	8.7	.11	8.5	.0
		5	53	8	68	0	78	92
	Control	12	5.7	.18	6.8	.14	6.6	.17
		5	03	3	52	2	38	5

EFC/C number of error-free clauses to total number of clauses

Furthermore, the pairwise analysis showed that students in the experimental group improved their accuracy in writing argumentative essays in the posttest and delayed posttest phases. However, students in the control group showed no significant improvement in accuracy across phases.

The intergroup analysis also found that both groups exhibited comparable accuracy in writing argumentative essays in the pretest phase. However, in the posttest and delayed posttest phases, the experimental group's accuracy improved significantly compared to the control group; with a p -value <0.001 , and a medium effect size, Cohen's $d = 0.573$.

Further analysis was conducted on the fluency aspect of students' argumentative essays. The analysis showed a significant improvement in argumentative essay fluency in the experimental group, but no improvement in fluency in the control group.

For a more detailed analysis, a mixed ANOVA test was conducted, the results of the mixed ANOVA test showed that there was a significant main effect of time with a value of $F(2, 60) =$

12.457, $p < 0.001$, η^2 Partial = 0.364 and a significant interaction effect of time and group with a value of $F(2, 60) = 4.786$, $p = 0.035$, η^2 Partial = 0.120. In addition, a significant main effect based on the group variable was also found on the fluency aspect with a value of $(1, 60) = 3.867$, $p = 0.054$, η^2 Partial = 0.054 on fluency performance.

Furthermore, pairwise analysis showed that collaborative writing intervention based on metacognitive instruction was able to improve the fluency performance of argumentative essay writing in the experimental group in each phase (pre- vs. post-test, $p < 0.05$; pre- vs. delayed post-test, $p < 0.001$). In addition, the intervention effect was also maintained in the delayed post-test phase (post- vs. delayed post-test, $p > 0.05$).

However, fluency performance in the control group did not show any improvement in all three phases ($p > 0.05$). Furthermore, based on the results of the inter-subject comparison analysis, fluency performance in both groups was found to be equivalent with a p value > 0.05 . However, in the post-test phase, fluency performance in the experimental group showed a better improvement than the control group with a p value = 0.001, with a large effect size, Cohen's $d = 37.563$. However, in the delayed post-test phase, no significant improvement was found. The results of the descriptive statistical analysis of argumentative essay writing fluency are presented in Table 4.

Table 4. Descriptive statistics of argumentative essay fluency.

Fluency	Group	N	Pre-test		Post-test		Delayed posttest	
			M	SD	M	SD	M	SD
Number of words	EG	125	5.648	.184	8.736	.241	8.786	.365
	CG	125	5.539	.268	6.425	.634	6.567	.645

The next step was a regression analysis to investigate which components of metacognitive instruction significantly contributed to argumentative essay writing skills. The results of the regression analysis are presented in Table 5.

The results of the regression analysis indicate that the eight components of metacognitive instruction used in collaborative writing accounted for 91% of the variance in scores on students' argumentative essay writing skills. This regression analysis confirms that the eight components of metacognitive instruction in collaborative writing significantly contributed to academic writing skills ($p < 0.001$).

Table 5. Results of linear regression.

	Unstandardized Coefficients		Standardized Coefficients	t	p	VIF	R ²	Adjusted R ²	F
	B	Std. E	Beta						

Consta nt	1.6 34	0.2 82	-	6 .5 7 3	0.0 00* *	-	0 .9 1	0.9 35	642. 43 ***
DK	0.0 97	0.0 18	0.147	8 .4 8 5	0.0 00* *	4. 3 7 8			
PK	0.1 25	0.0 22	0.15	7. 6 8 4	0.0 00* *	4. 7 5 2			
CK	0.1 36	0.0 27	0.142	7. 4 8 3	0.0 00* *	5. 6 3 5			
P	0.0 84	0.0 17	0.15	7. 8 5 6	0.0 00* *	5. 2 5 6			
M	0.0 98	0.0 19	0.172	7. 7 8 2	0.0 00* *	5. 4 7 8			
E	0.0 92	0.0 16	0.178	8 .6 9 4	0.0 00* *	5. 8 4 5			
IMS	0.0 93	0.0 17	0.15	7. 8 0 5	0.0 00* *	3. 7 6 2			
DS	0.0 93	0.0 16	0.152	8 .3 7 8	0.0 00* *	3. 5 4 8			

Declarative Knowledge (DK), Procedural Knowledge (PK), Conditional Knowledge (CK), Planning (P), Monitoring (M), Evaluation (E), Information Management Strategy (IMS), and Debugging Strategy (DS).

5. Discussion

This study aimed to investigate the impact of integrating Automatic Corrective Feedback (ACF) into collaborative writing based on metacognitive instruction on argumentative essay writing skills. The research findings indicate that the intervention of integrating Automatic Corrective Feedback (ACF) into collaborative writing based on metacognitive instruction can improve argumentative essay writing skills. Improvements in students' writing skills were evident in the quality of their argumentative essays, including lexical complexity (lexical density, sophistication, and variety), accuracy, and fluency.

The increase in lexical complexity was evident in the density, sophistication, and variety of

lexical items, which were quite significant as a result of the intervention. The increase in lexical complexity in the experimental group was greater than in the control group, which received the genre-based writing intervention.

The increase in lexical complexity occurred due to the integration of metacognitive instruction in collaborative writing, which encompassed a combination of metacognitive knowledge (declarative, procedural, and conditional knowledge), which facilitated students' understanding of the concept and procedures for writing argumentative essays. Furthermore, metacognitive regulation (planning, monitoring, and evaluation), as well as information management and debugging strategies, included in the metacognitive instruction, improved argumentative essay writing skills from the planning stage to revision. This finding aligns with previous studies showing that metacognitive strategy-based writing instruction facilitates students' improvement in academic writing skills and self-evaluation (Paulsen & Davidsen, 2025; Mekheimer, 2025).

Furthermore, other studies have shown that collaborative writing improves essay quality because learners can work collaboratively and receive feedback during the writing process (Khuder & Negretti, 2025; Sun & Zhang, 2023). This process is believed to facilitate student writing and improve the quality of their writing.

Another discovery was that the GenAI-ACF integration intervention in collaborative writing using metacognitive instruction enhanced the precision of argumentative essays in the experimental group, surpassing the control group, which underwent a genre-focused writing intervention. This improvement in precision was apparent at all stages, from pretest to posttest and posttest to delayed posttest, whereas the control group demonstrated steady or unchanging values.

The rise in precision in the delayed posttest shows that the intervention for the experimental group had a lasting effect. Accuracy in writing argumentative essays was demonstrated through the coherence and cohesion of the text both among sentences and within paragraphs.

Additionally, precision was also apparent in backing and emphasizing concepts related to the issue of the argumentative essay. This increase in accuracy occurred due to the metacognitive knowledge stage that improved students' understanding of argumentative essay writing and metacognitive regulation facilitated by GenAI-ACF, which facilitated students in planning, monitoring, and evaluating their writing well.

The evaluation stage from GenAI-ACF and peers provided students with the opportunity to reread their work and correct any errors. This process was deemed capable of improving the accuracy of argumentative essay writing. Furthermore, other findings confirmed that metacognitive instruction in collaborative writing can improve writing accuracy, especially in the accuracy of essay topic development and using vocabulary, sentences in the context (Anggraeni et al., 2025; Mekheimer, 2025).

Other studies also showed that collective writing in pairs showed advantages in accuracy aspects compared to individual writing (Feng et al., 2024; Kim et al., 2024). This finding is also in line with the collaborative learning theory that collaborative writing facilitates students in building content knowledge and language use through collaborative discussions so that learners are able to produce sentences with more accurate grammar (Rahimi et al., 2024; ten Peze et al., 2024).

The next finding was that the integration of GenAI-ACF in collaborative writing based on metacognitive instruction improved argumentative essay fluency better than genre-based instruction. Improvements in fluency in the experimental group were evident in the posttest and maintained through the delayed posttest. Improved fluency in argumentative essays was evident in language use, organization, and smooth transitions between paragraphs without sentence errors.

This improvement occurred because the GenAI-ACF platform and metacognitive instruction facilitated the effective transfer of strategies for planning, monitoring, and evaluating, allowing for correction of sentence errors. Furthermore, collaborative writing also contributed to argumentative essay fluency because it provided opportunities for more varied language use, provided a sharing phase, and strengthened arguments through feedback (Khosravi et al., 2023; Mohammed & Khalid, 2025). These findings align with previous studies that confirm that collaborative writing can increase the variety of language and argument used in essays (Janebi Enayat, 2025; Jin, 2025).

Several aspects of the collaborative writing process using GenAI-ACF based on metacognitive instruction significantly contributed to writing ability. Contributions to conceptual mastery stem from metacognitive knowledge aspects, including declarative, procedural, and conditional knowledge, which help students comprehensively understand the concept of argumentative essays, from the concept definition to the procedure for creating them.

Furthermore, metacognitive regulation aspects, including planning, monitoring, and evaluation, as well as information management and debugging strategies, facilitate students in creating structured and evaluative argumentative essays because metacognitive instruction accommodates everything from the planning stage to the revision of argumentative essay writing. This process contributes significantly to improving the quality of argumentative essays. This finding aligns with metacognitive theory, which states that individual metacognitive abilities significantly determine how students write and improve their writing (Carmona et al., 2025; Mekheimer, 2025).

This finding is also supported by previous studies confirming that metacognitive strategies not only improve students' scientific writing skills but also enhance their self-evaluation skills in improving the quality of their writing (Anggraeni et al., 2025; Rezai et al., 2023). Other studies indicate that metacognitive regulation is a significant contributor to students' writing styles and writing quality because this component is already included in the application stage of writing comprehension in producing a piece of writing (Kim et al., 2024; Rahimi et al., 2024).

6. Conclusion

The integration of GenAI-ACF in collaborative writing based on metacognitive instruction significantly improved argumentative essay writing skills compared to genre-based writing interventions. Improvements in argumentative essay writing skills in the experimental group were evident in several aspects, including lexical complexity (lexical density, lexical sophistication, and lexical variety), accuracy, and fluency. Improved lexical complexity performance was evident in the use of complex vocabulary, complex sentence combinations, and more varied lexical use. Improved accuracy in argumentative essays was evident in the accuracy of inter-sentence and paragraph coherence.

Furthermore, accuracy was also evident in the use of supporting and reinforcing ideas relevant to the problem of the argumentative essay. Improved fluency in argumentative essays was evident in the use of sentences and grammar with minimal errors and no discordant sentences. Improvements in all aspects of lexical complexity, accuracy, and fluency occurred because the collaborative writing components and metacognitive instruction significantly contributed to argumentative essay writing skills. Several collaborative writing components that contributed significantly were the distribution of writing tasks, collaborative discussions, and feedback.

A significant component contributing to argumentative essay writing skills is metacognitive knowledge, encompassing declarative, procedural, and conditional knowledge, which helps students comprehensively understand argumentative essay concepts, from definitions to the procedures for writing them. The regulatory component of metacognition, encompassing

planning, monitoring, and evaluation, as well as information management and debugging strategies, facilitates students in creating structured and evaluative argumentative essays because it encompasses the planning and revision stages of argumentative essay writing.

This research suggests that the integration of GenAI-ACF in collaborative writing and metacognitive instruction can enhance learner active participation and improve writing skills. Furthermore, this intervention not only improves writing skills but also enhances the ability to evaluate one's own writing. Increasing students' metacognitive awareness can effectively facilitate students' improvement in several dimensions of writing performance, thereby improving writing quality.

This research highlights the significance of combining GenAI technology, cooperative writing, and metacognitive instruction to enhance the growth of students' writing abilities. This study enhances the comprehension of collaborative writing tasks as a valuable teaching method and enriches the understanding of metacognitive knowledge processes, metacognitive regulation, and the intricate writing dynamics involved in collaborative writing.

This research enhances the metacognitive viewpoint in collaborative writing, promoting additional investigations into metacognitive processes related to collaborative writing and its mechanisms. This research has various constraints, such as the brief duration of the intervention, the absence of an in-depth examination of the collaborative writing process, a small sample size, and the omission of qualitative analysis of the outcomes of the intervention.

Given these restrictions, the researcher proposes various recommendations for upcoming studies, such as extending the intervention duration if needed, conducting a longitudinal study for improved impact, analyzing the collaborative writing process along with the dialogue utilized, increasing the sample size concerning both levels and numbers, and performing a qualitative analysis to uncover more thorough insights for achieving better outcomes in writing argumentative essays.

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